

Questions: Creative Action

Analytical Questions

1. Explain how creative action functions as both pragmatic inference (changing the world to match the model) and epistemic inference (generating observations to update the model). Why is the epistemic function especially important in the early stages of invention?
2. James Dyson built 5,127 prototypes of the bagless vacuum. Analyze this as an Active Inference process. What role did each prototype play in the perception-action loop, and why is "5,127 failures" a misleading description?
3. Define fidelity matching in prototyping. Why does building a high-fidelity prototype to answer a question that a low-fidelity prototype could answer actually produce worse information, not just wasted resources?
4. How does sketch modeling engage cognitive resources that mental simulation cannot access? Use the concept of embodied cognition to explain why manipulating physical objects provides different information than imagining them.
5. Analyze the design sprint methodology as a compressed Active Inference cycle. Map each of the five days (Map, Sketch, Decide, Prototype, Test) to the corresponding Active Inference operation and explain why the time constraint is a feature, not a limitation.
6. Define bricolage in Active Inference terms. How does the bricoleur's relationship between model and world differ from the conventional engineer's relationship, and under what conditions is bricolage the superior strategy?
7. The Palm Pilot was developed using a carved wooden block as a prototype. Analyze this in terms of the perception-action loop and fidelity matching. What specific types of information did the wooden block provide that no other method (mental simulation, computer modeling, verbal description) could have provided?

8. Compare the Apollo 13 CO₂ scrubber improvisation with a conventional engineering design process. How did the extreme constraint (only onboard materials) change the relationship between the generative model and the available actions?
9. Research in design cognition shows that early physical prototyping reduces overall project time and cost. Explain this counterintuitive result using Active Inference: why does spending time on action (building) reduce the total time spent on cognition (thinking)?
10. The concept of "minimum viable prototype" prioritizes testing the most critical assumption first. How does this relate to the Active Inference concept of expected free energy? What makes an assumption "most critical" in information-theoretic terms?

Applied Questions

1. Identify the single most critical uncertainty about your invention idea. Design a prototype that tests this uncertainty at the lowest possible fidelity. Describe what you would build, what materials you would use, and what observation would confirm or refute your assumption.
2. Build a five-minute prototype of your invention using only materials within arm's reach. Document what you built, what surprised you, and what prediction errors occurred during construction. What did making reveal that thinking had not?
3. Describe a situation where you spent too long thinking about an invention before building anything. What was the cost of delayed action? What would have happened if you had built a rough prototype on day one?
4. Apply the bricolage approach to your invention: list ten materials currently available to you and describe how each could be repurposed for prototyping a component of your invention. Which material-use pairings surprise you?
5. Design a one-day design sprint for your invention project. Map out the five activities (Map, Sketch, Decide, Prototype, Test) with time allocations and describe what you would produce at each stage.

6. Take your current best invention concept and identify three aspects that require different prototype fidelities to test. For each, specify the fidelity level, the materials needed, and the specific question the prototype answers.
7. Show a sketch or prototype of your invention to someone unfamiliar with it. Document their reaction, their questions, and any usage behavior you did not expect. What prediction errors did observing them generate?
8. Describe the most instructive prototype failure you have experienced (or can imagine for your current project). What did the failure teach you that success would not have? How did the failure update your generative model?
9. Compare making your prototype with your hands (physical materials) versus describing it in words versus drawing it on paper. Which externalization method provides the richest information? Why?
10. Design a "prototype roadmap" for your invention: a sequence of 3-5 prototypes at increasing fidelity levels, each testing a different critical assumption. For each prototype, specify the question, the fidelity, the materials, and the success/failure criteria.